

The Similarity Toolkit — A Reference Guide

Congruence made copies; similarity makes scale models. This page maps the new tools onto the old ones.

	Congruence (Ch. 5)	Similarity (Ch. 8)
What matches	sides congruent, angles congruent	angles congruent, sides PROPORTIONAL
Symbol	$\cong$	$\sim$
Size	identical	same shape, any size — measured by the scale factor
Triangle shortcuts	SSS · SAS · ASA · AAS · HL	AA · SSS~ (ratios) · SAS~ (ratios + included angle)
Why AA has no congruence twin	AAA is NOT a congruence criterion — size is free	two angles are ENOUGH for similarity, since shape is all that matters
After you prove it	CPCTC transfers congruent parts	corresponding angles are congruent; corresponding sides share one ratio

Setting Up a Proportion Without Getting Burned

- 1. Write the similarity statement FIRST and read corresponding parts from position: in $\triangle JKL \sim \triangle PQR$, side $\overline{JK}$ matches $\overline{PQ}$ because $J \leftrightarrow P$ and $K \leftrightarrow Q$.
- 2. Keep one figure on top. 'Big over small' everywhere, or 'small over big' everywhere — never mixed in one equation.
- 3. The scale factor has a direction. $\triangle ABC$ to $\triangle DEF$ may be $3/2$ while $\triangle DEF$ to $\triangle ABC$ is $2/3$. Say which way before you multiply.
- 4. Perimeter scales like the sides — one scale factor for both. (Area does not; that is a later chapter.)
- 5. Cross-multiply LAST, after the proportion is built from corresponding parts.

The 8.4 Picture Dictionary

Three theorems, three pictures. Match the picture, and the proportion writes itself.

One Parallel Line Inside a Triangle (Triangle Proportionality)

- A segment parallel to one side splits the other two sides in the same ratio: top piece / bottom piece = top piece / bottom piece.
- The whole triangle and the small one are also similar by AA — use whole sides ($BX/BA = BY/BC$) only in THAT proportion; never mix pieces with wholes.
- Converse: equal ratios prove the segment parallel. Compute both ratios and compare — that is the entire test.

Three Parallel Lines Across Two Transversals

- The parallels slice both transversals in the same ratio: left-top/left-bottom = right-top/right-bottom.
- The transversals themselves need not be parallel, and the pieces on one transversal can be longer than the other — only the RATIO carries across.

An Angle Bisector of a Triangle

- The bisector from a vertex splits the OPPOSITE side in the ratio of the two ADJACENT sides: near piece / far piece = near side / far side.
- Fastest solve: call the pieces $3k$ and $2k$ (or whatever the ratio is), add them to the whole side, and solve for k .
- Sanity check: the longer piece of the opposite side always sits next to the longer adjacent side.

Indirect Measurement in One Line

- Same sun, same moment → similar triangles by AA. height / height = shadow / shadow. One unknown, one cross-multiplication.

Name: _____

Which Tool? Practice

Name the postulate or theorem that applies — AA, SSS Similarity, SAS Similarity, Triangle Proportionality (or its Converse), Three Parallel Lines, Angle Bisector — or NOT ENOUGH.

1. Two pairs of congruent angles.
2. All three pairs of sides in the same ratio.
3. Two pairs of proportional sides and congruent included angles.
4. Two pairs of proportional sides and a congruent angle NOT between them.
5. A segment inside a triangle is parallel to one side; you need a missing piece of a side.
6. You know four segment pieces and must DECIDE whether the segment is parallel.
7. Three parallel streets cross two straight avenues.
8. A ray splits a triangle's angle in half; the opposite side is cut into two pieces.
9. A person's shadow and a tower's shadow at the same time of day.
10. Two equilateral triangles of different sizes.
11. Both triangles contain a 90° angle and share an acute angle.
12. The three ratios compute to $\frac{2}{3}$, $\frac{2}{3}$, and $\frac{3}{4}$.
13. A midsegment of a triangle.
14. Two rectangles.
15. Two squares.